Acute Inflammation

Complete study notes: Robbins lecture note + Flipped case (Darla) + Practical
MDP10906 Paraclinical Sciences · 2026/2027 · Prof. Dr. Dayangku Norlida

What this is: a rewrite of all three slide decks in simpler language, keeping every topic in the slides (only memes, jokes and decorative pictures are left out). Slide figures and practical pictures are included, so you do not need to open the slides to follow along. Yellow boxes explain things in plain words.

The 7 learning objectives (and where to find them)

Objective	Section
1. Define and state the aim of inflammation	1
2. Five cardinal signs and their pathophysiology	2
3. Three components: vascular events, cellular events, chemical mediators	6, 8, 9, 10, 12
4. Causes of acute inflammation	3
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6. Outcomes of acute inflammation	14
7. Morphological patterns of acute inflammation	13
Practical (cells, pneumonia, appendicitis) / Darla case	Part 2 / Part 3

PART 1 · THEORY

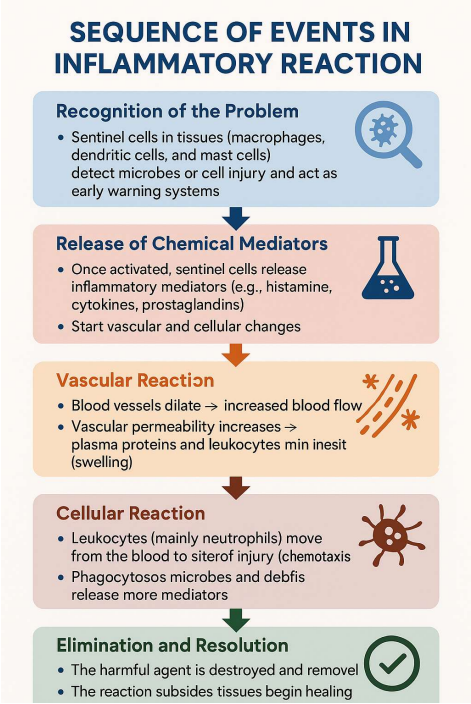
1. Definition, aim and big picture

- **Inflammation** = the protective response of vascularised tissue that brings white blood cells and defence molecules (antibodies, complement) from the blood to the site of infection or injury, to eliminate the harmful cause.
- **Aim:** remove the cause (microbes, toxins), remove the damaged tissue it caused, and start repair. It looks harmful (swelling, pain) but it is essential: without it infections spread unchecked, wounds do not heal and tissue stays damaged.
- **Key players:** leukocytes/phagocytes (kill pathogens, clear debris), antibodies and complement proteins (strengthen defence), and tissue sentinel cells (detect threats early).
- **Process:** blood flows through all tissues, so immune cells and proteins can be delivered quickly to the damaged site, where they become activated.
- Terminology: the suffix **-itis** means inflammation (appendicitis, conjunctivitis, meningitis).

Historical highlights

- Signs of inflammation were recorded in Egyptian papyrus (about 3000 BC). **Celsus** (1st century) gave 4 cardinal signs: rubor, tumor, calor, dolor. **Virchow** (19th century) added the 5th: functio laesa.
- **John Hunter** (1793): inflammation is not a disease but a protective response that helps the body heal.
- **Elie Metchnikoff** (1880s): discovered phagocytosis, showing the main purpose of inflammation is to bring phagocytes to engulf microbes and debris.

The 5 steps (Rs) of an inflammatory reaction



Sequence of events in the inflammatory reaction (slide infographic)

Step	What happens
1. Recognition	Sentinel cells (macrophages, dendritic cells, mast cells) and phagocytes have receptors that detect harmful agents (microbes, toxins, cell injury). When activated they release inflammatory mediators, which trigger the next steps.
2. Recruitment	Leukocytes (neutrophils, monocytes, lymphocytes) and plasma proteins (complement) move from the blood to the injury site. This needs changes in blood vessels and mediators so cells and proteins can leave the circulation.
3. Removal	Phagocytes (mainly macrophages and neutrophils) ingest and destroy microbes, dead cells and debris. This clears the cause and prepares for healing.
4. Regulation	The reaction must be controlled and stopped once the stimulus is gone; anti-inflammatory mediators switch it off to prevent excess tissue damage.
5. Repair	Healing starts after inflammation subsides: regeneration of surviving cells and scar formation (fibrosis) to fill remaining defects.

Basic components and harmful consequences

- **Blood vessels:** dilate (more blood flow = redness and heat); become more permeable (plasma proteins enter tissue); endothelial cells change so leukocytes can stick and leave the circulation.
- **Leukocytes:** once in tissue they become activated, phagocytose and destroy microbes, dead cells and foreign material.
- **Harmful side:** normal inflammation may cause pain, swelling and temporary loss of function but usually heals. When misdirected or uncontrolled it causes disease: *autoimmune* (targets own tissue, e.g. rheumatoid arthritis), *allergic* (harmless substances: pollen, food, insect stings, drugs) and *chronic* inflammation (atherosclerosis, lung fibrosis, type 2 diabetes, Alzheimer disease, cancer; called "the silent killer").
- Anti-inflammatory drugs aim to limit tissue injury while keeping the benefits of inflammation.

Local vs systemic; mediators in brief

- **Local inflammation:** at one site (cut, abscess); local redness, swelling, pain, sometimes mild fever.
- **Systemic inflammation:** spreads through the body, e.g. severe infection/sepsis; can cause organ failure. Called **SIRS** (systemic inflammatory response syndrome).
- **Mediators** are soluble chemicals released by cells (macrophages, mast cells, endothelial cells) or from plasma proteins (complement). Triggers: microbes, dead or injured cells, hypoxia. They start, amplify and control the response, deciding its pattern, intensity and outcome.

Acute vs chronic inflammation

Feature	Acute	Chronic
Onset	Fast: minutes or hours (lasts hours to a few days)	Slow: days or longer
Cellular infiltrate	Mainly **neutrophils**	Monocytes/macrophages and lymphocytes

Feature	Acute	Chronic
Tissue injury, fibrosis	Usually mild and self-limited	Often severe and progressive
Local and systemic signs	Prominent	Less
Cause	Rapid response to an easily removed cause (bacteria, fungi, dead cells). Hallmarks: fluid and protein leakage (oedema) + leukocyte migration, mainly neutrophils. Subsides and heals once cause is removed.	Acute fails to resolve, or starts de novo. Persistent or hard-to-remove agent (TB bacilli, viruses, fungi, self-antigens). Ongoing tissue destruction, scarring and fibrosis. Can coexist with acute (e.g. peptic ulcer).

2. The five cardinal signs

Rubor (redness) · Tumor (swelling) · Calor (heat) · Dolor (pain) · Functio laesa (loss of function)

Sign	Pathophysiology
Redness (rubor) and heat (calor)	**Histamine** relaxes arterioles (precapillary sphincters) → more blood flows into capillaries and venules, which dilate → redness and heat at the inflamed area.
Swelling (tumor)	Histamine increases permeability of **venules** → fluid and proteins leak into surrounding tissue → **oedema** (excess fluid in the interstitial space).
Pain (dolor)	**Prostaglandin E2 (PGE2)** makes nerve endings more sensitive to **bradykinin** and other pain mediators, causing the pain of inflammation.
Loss of function (functio laesa)	Follows from pain and tissue injury (and swelling).

3. Causes of acute inflammation

Cause	Details
1. Infections	Most common and most important cause. Bacteria, viruses, fungi, parasites, microbial toxins. Outcome ranges from mild acute (heals fully), to severe systemic (may be fatal), to chronic (tissue damage); depends on pathogen, host immune response and host factors.
2. Tissue necrosis	Any cell death can trigger it: ischaemia (myocardial infarction), trauma, burns, frostbite, radiation, chemical injury. Molecules released from dead cells act as signals.
3. Foreign bodies	Splinters, dirt, sutures (directly or through injury or microbial contamination). Also endogenous substances deposited abnormally: urate crystals (gout), cholesterol crystals (atherosclerosis), lipids (obesity-related inflammation).
4. Immune reactions	Immune system harms own tissue: autoimmune disease (rheumatoid arthritis), allergy (pollen, food), excessive response to microbes. These are chronic and hard to cure because the stimulus persists.

4. Recognition of microbes and damaged cells

Recognition is the initiating step of inflammation.

Recognition system	Details
Cellular receptors for microbes	Plasma membrane (extracellular microbes), endosomes (ingested microbes), cytosol (intracellular microbes). Most important = **Toll-like receptors (TLRs)** on epithelial cells, macrophages, dendritic cells and leukocytes; activation releases mediators (cytokines, chemokines).
Sensors of cell damage	All cells have cytosolic **NOD-like receptors (NLRs)**. They detect uric acid (from DNA breakdown), ATP (from damaged mitochondria), low intracellular K^+ (membrane injury) and cytoplasmic DNA. This activates the **inflammasome**, which releases **IL-1** to recruit leukocytes and start inflammation.
Other receptors	Leukocytes recognise microbes coated with antibodies or complement (**opsonisation**), which helps phagocytosis and amplifies inflammation.
Circulating proteins	**Complement** (reacts against microbes, generates mediators); **mannose-binding lectin (MBL)** (binds microbial sugars, aids phagocytosis, activates complement); **collectins** (bind and neutralise microbes).

5. The three components of acute inflammation

- **(1) Dilation of small vessels** → more blood flow → redness and heat.
- **(2) Increased vascular permeability** → plasma proteins and leukocytes leak out → oedema.
- **(3) Emigration of leukocytes** → WBCs leave the circulation, accumulate at the site and are activated to destroy the harmful agent.

- **Initiation:** when a microbe or dead cell is detected, resident phagocytes (macrophages, dendritic cells, mast cells) try to eliminate it. They release cytokines and lipid mediators that act on nearby vessels and cause fluid and leukocyte movement into the tissue.
- Grouping: **vascular events** (1 and 2) and **cellular events** (3), all driven by **chemical mediators**.

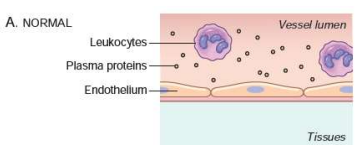
6. Vascular events (reactions of blood vessels)

Sequence

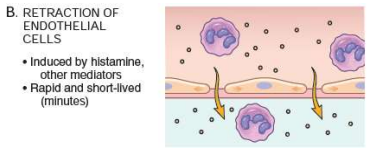
Step	What happens
1	Immediate, brief **vasoconstriction** of arterioles (neurogenic reflex, a few seconds).
2	Persistent **vasodilation** of arterioles (histamine, prostaglandins, kinins, NO) and opening of new capillary beds → local heat and redness.
3	**Increased vascular permeability** (endothelial contraction or injury) = hallmark of acute inflammation.
4	Exudation of fluid into extracellular space = **oedema**.
5	Fluid loss + dilation → slower flow, red cell concentration, higher viscosity → vascular congestion and **stasis**.
6	During stasis, **margination**: neutrophils move to the vessel edge (RBCs are smaller, faster and stay central) and endothelium expresses adhesion molecules, so leukocytes stick and then migrate into the tissue.

Mechanisms of increased permeability (postcapillary venules)

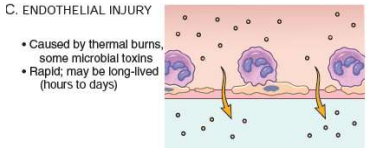
Mechanism	Features
Contraction of endothelial cells	Most common. Histamine, bradykinin, leukotrienes → gaps between endothelial cells. **Immediate transient response**: within minutes, lasts 15–30 min. **Delayed prolonged leakage**: mild injury (burns, UV, radiation, bacterial toxins); begins 2–12 h later, lasts hours to days; sunburn is the classic example. Both can occur in sequence.
Endothelial injury	Direct damage causing necrosis and detachment: severe injury (thermal burns) or microbial toxins. Neutrophils stuck to endothelium add secondary injury. Leakage starts immediately and lasts hours until vessels are thrombosed or repaired. Mechanisms often act together (burn: mediators + direct injury + neutrophil damage = major, possibly life-threatening fluid loss).



A. Normal

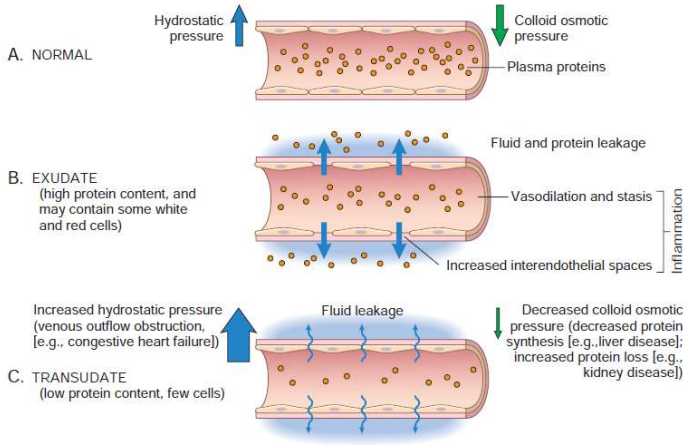


B. Endothelial retraction (histamine; minutes)



C. Endothelial injury (burns; hours to days)

Exudate, transudate, oedema, pus



Fluid movement across capillaries: (A) normal, (B) exudate, (C) transudate

Term	Meaning
Exudate	High protein content and cellular debris. Increased vascular permeability = inflammation.
Transudate	Low protein (mostly albumin), few or no cells. Osmotic or hydrostatic imbalance, **not** inflammation.
Oedema	Excess fluid in tissues or body cavities; can be exudate or transudate. (Fluid in a body cavity is also called an **effusion**.)
Pus (purulent exudate)	Inflammatory exudate rich in neutrophils, dead cells and often microbes.

Normal pressures: hydrostatic ~32 mmHg (arterial end) to ~12 mmHg (venous end); colloid osmotic ~25 mmHg. They balance, so there is almost no net fluid movement. Exudate forms in inflammation (↑ permeability, wider endothelial gaps). Transudate forms when hydrostatic pressure rises (heart failure) or osmotic pressure falls (low plasma proteins).

Lymphatics and lymph nodes

- Lymphatics normally drain the small amount of fluid that escapes capillaries. In inflammation, lymph flow increases to remove oedema fluid, leukocytes, debris and microbes, and lymphatic vessels proliferate.
- Draining nodes enlarge (hyperplasia of lymphoid follicles, more lymphocytes and macrophages) = **reactive (inflammatory) lymphadenitis**.
- Clinical: red streaks near a wound = inflamed lymphatic channels = **lymphangitis**; painful enlarged nodes = **lymphadenitis**.



Lymphangitis (red streak)



Lymphadenitis (swollen neck node)

7. Oedema: aetiology and pathogenesis

- **Oedema** = excess fluid in tissues; **effusion** = excess fluid in body cavities (pleura, pericardium, peritoneum). Either can be inflammatory (exudate) or non-inflammatory (transudate).
- **Normal balance:** hydrostatic pressure pushes fluid *out* of capillaries; colloid osmotic pressure (plasma proteins) pulls fluid *back in*; lymphatics drain the small excess back to the blood. Oedema develops when more fluid leaves than is drained.

Category	Mechanism	Examples
1. ↑ Hydrostatic pressure	Impaired venous return pushes more fluid into tissue	Heart failure, DVT; also constrictive pericarditis, ascites (cirrhosis), venous compression, prolonged standing; arteriolar dilation (heat)
2. ↓ Plasma osmotic pressure	Low albumin (↓ synthesis or ↑ loss) → less fluid drawn back	Liver disease/cirrhosis (↓ synthesis); nephrotic syndrome, protein-losing enteropathy (↑ loss); malnutrition
3. Na^+ and water retention	↑ blood volume (↑ hydrostatic) and dilution (↓ osmotic)	Renal dysfunction, excess salt intake, ↑ renin-angiotensin-aldosterone
4. Lymphatic obstruction	Fluid cannot drain → lymphoedema	Trauma, fibrosis, tumours, infection, surgery, radiation
5. Inflammation	↑ vascular permeability → protein-rich exudate	Acute and chronic inflammation, angiogenesis

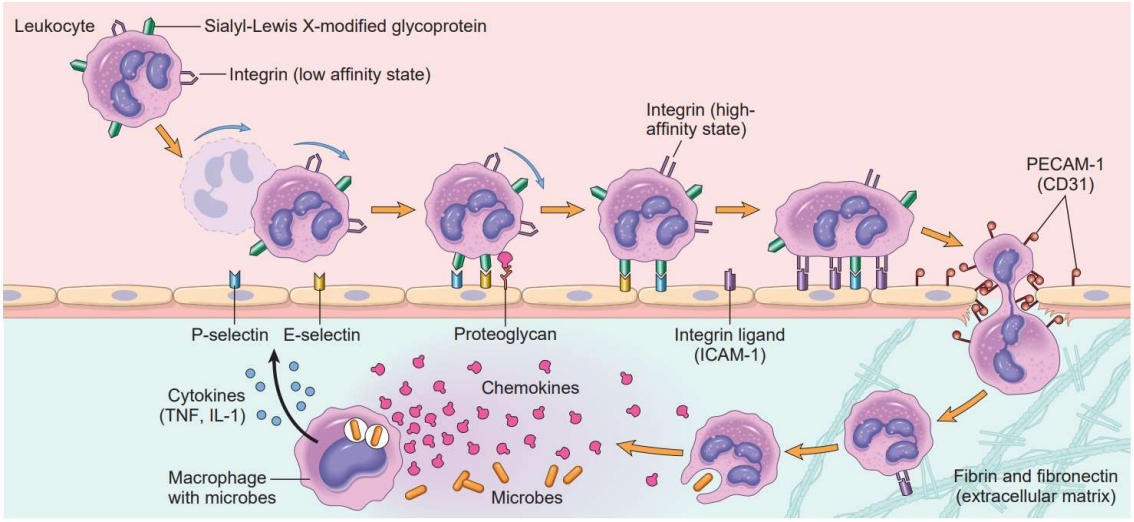
Non-inflammatory (transudate) oedema: protein-poor; imbalance in hydrostatic/osmotic pressure; heart, liver or renal failure, malnutrition. **Inflammatory (exudate)**: protein-rich; contains cells, proteins and mediators; from ↑ permeability (e.g. pus).

Increased Hydrostatic Pressure
Impaired Venous Return
Congestive heart failure Constrictive pericarditis Ascites (liver cirrhosis) Venous obstruction or compression Thrombosis External pressure (e.g., mass) Lower extremity inactivity with prolonged dependency
Arteriolar Dilation
Heat Neurohumoral dysregulation
Reduced Plasma Osmotic Pressure (Hypoproteinemia)
Protein-losing glomerulopathies (nephrotic syndrome) Liver cirrhosis Malnutrition Protein-losing gastroenteropathy
Lymphatic Obstruction
Inflammatory Neoplastic Postsurgical Postirradiation
Sodium Retention
Excessive salt intake with renal insufficiency Increased tubular reabsorption of sodium Renal hypoperfusion Increased renin-angiotensin-aldosterone secretion
Inflammation
Acute inflammation Chronic inflammation Angiogenesis

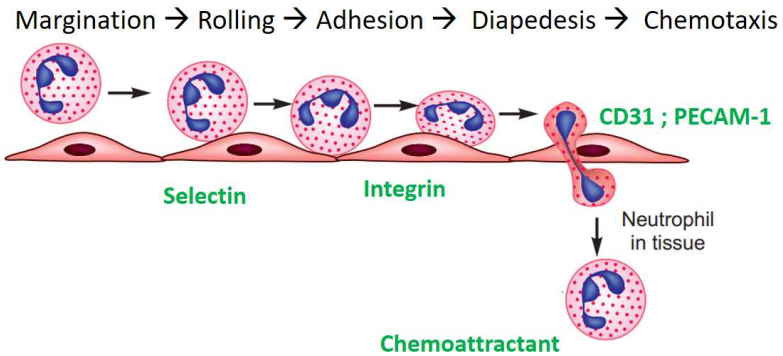
Causes of oedema (slide table)

8. Cellular events: leukocyte recruitment

After the vascular changes there is a rapid influx of leukocytes whose job is to eliminate the offending agent. Neutrophils and macrophages are the main phagocytes: they ingest microbes, necrotic tissue and foreign material, and macrophages also release growth factors for repair. **Collateral damage:** activated leukocytes release enzymes and reactive substances that kill microbes but can also injure normal tissue and prolong inflammation.



The multistep process of leukocyte migration through blood vessels (neutrophils)

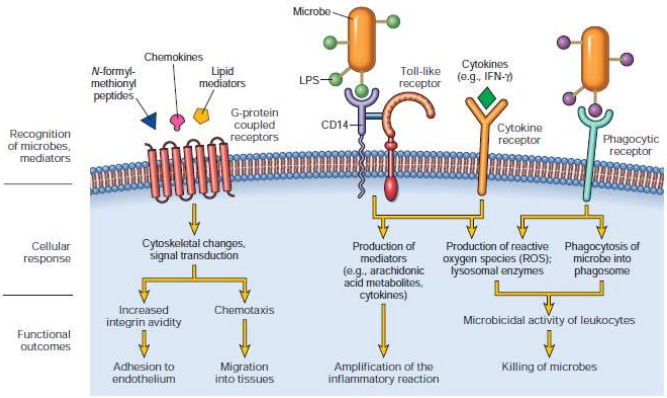


Margination → rolling (selectins) → adhesion (integrins) → diapedesis (CD31/PECAM-1) → chemotaxis

Phase	What happens	Key molecules
In the lumen: Margination	Normally RBCs flow centrally and push heavier, slower WBCs to the wall. In inflammation, vasodilation slows flow (stasis), so more leukocytes move to the periphery.	Stasis

Phase	What happens	Key molecules
Rolling	Slowed leukocytes sense signals from activated endothelium and roll along it (like pebbles on a streambed). Normal endothelium does not bind leukocytes.	**Selectins**: L-selectin (leukocytes), E-selectin (endothelium), P-selectin (platelets and endothelium). Cytokines raise selectin expression.
Adhesion (firm)	After rolling, leukocytes slow down and bind firmly, anchoring before migration.	**Integrins** on leukocytes bind ligands on endothelium (ICAM-1, VCAM-1) induced by TNF and IL-1; chemokines activate integrins.
Transmigration (diapedesis)	Cross between endothelial cells, mainly in postcapillary venules, guided by chemokines (chemotactic gradient); pierce basement membrane, which reseals afterwards.	CD31 (PECAM-1); collagenases secreted by leukocytes
Chemotaxis	Movement along a chemical gradient toward the injury.	Exogenous: bacterial products. Endogenous: chemokines (IL-8), complement C5a, leukotriene LTB4

Leukocyte sequence and lifespan	Details
Neutrophils	Dominate in first 6–24 h. Early dominance because they are more numerous in blood, respond faster to chemokines and bind P-/E-selectin more strongly. Short-lived: apoptosis within a few days.
Monocytes/macrophages	Replace neutrophils at 24–48 h. Survive longer, proliferate in tissue, dominate in prolonged inflammation. Cell type can vary with stimulus and stage.
Leukocyte activation	Neutrophils and monocytes must be activated at the site via pattern-recognition receptors (TLRs) → phagocytosis and destruction. Steps: recognition → activation → defence responses. Examples: LPS binds LPS-binding protein → activates leukocytes; IFN-γ activates macrophages → better microbe killing.

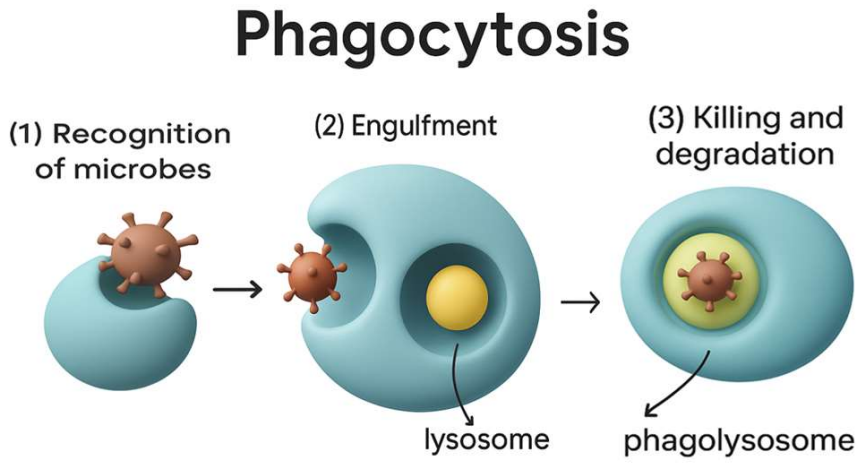


Leukocyte activation: receptors → signalling → functional responses (phagocytosis, killing, chemotaxis, amplification)

In plain words: WBCs normally hurry down the middle of the vessel. In inflammation the vessel dilates and slows, so WBCs drift to the edge (margination), roll on selectins, stick tightly with integrins, squeeze out between endothelial cells (diapedesis) and follow chemical signals (chemotaxis) to the injury. Neutrophils get there first.

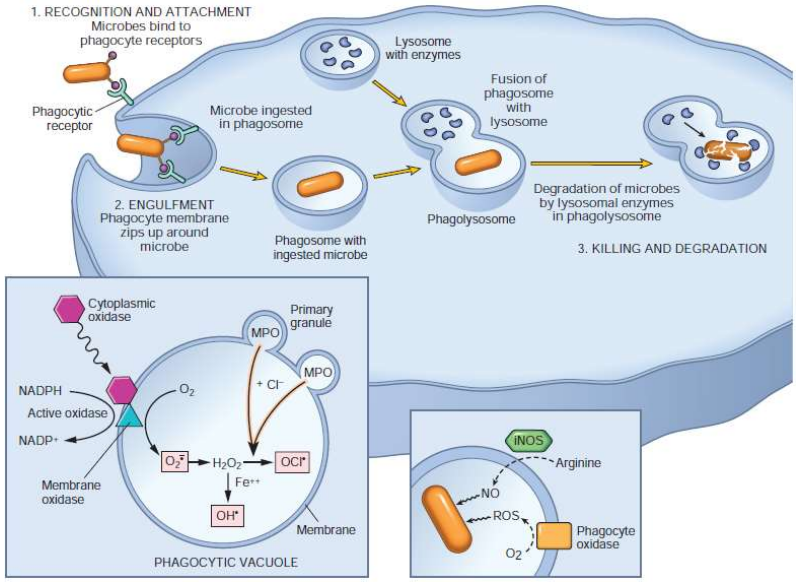
9. Phagocytosis and clearance of the offending agent

Neutrophils and macrophages are the two main phagocytes. Recognition of microbes/dead cells activates them (↑ cytosolic Ca^{2+}, activation of enzymes such as protein kinase C and phospholipase A_2). Two main responses: phagocytosis and intracellular killing.



Phagocytosis: (1) recognition, (2) engulfment, (3) killing and degradation

Step	Details
1. Recognition and attachment	Phagocyte surface receptors bind the microbe: **mannose receptors** (mannose/fucose on microbial walls, not host cells) and **scavenger receptors** (modified LDL, many microbes). **Opsonins** coat microbes to enhance phagocytosis; phagocytes have high-affinity receptors for them. Major opsonins: **IgG, C3b, mannose-binding lectin and collectins**.
2. Engulfment	Cytoplasm extends around the particle; membrane encloses it forming a **phagosome**, which fuses with a **lysosome** forming a **phagolysosome** where digestion occurs. Some lysosomal contents leak outside the cell and add to tissue damage.
3. Killing and degradation	Final step; inside activated neutrophils and macrophages. **Oxygen-dependent:** myeloperoxidase (MPO) system and reactive oxygen species (ROS). **Oxygen-independent:** lysosomal enzymes and reactive nitrogen species (mainly nitric oxide, NO). All killing is contained in the lysosome to protect cytoplasm and nucleus.

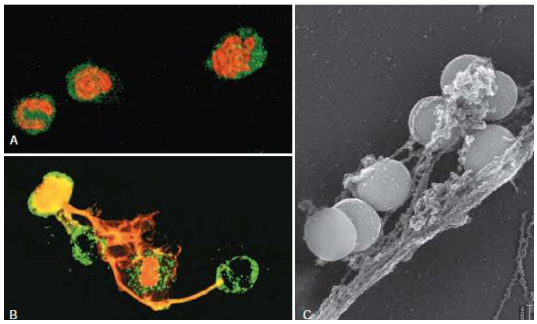


Phagocytosis and intracellular destruction: NADPH oxidase makes ROS; MPO converts H_2O_2 to hypochlorite (HOCl); iNOS makes NO → peroxynitrite

Killing summary: phagosome + lysosome → phagolysosome; enzymes digest; ROS and NO kill; MPO forms hypochlorite (HOCl); hydroxyl radical (·OH) and peroxynitrite ($OONO^-$) also destroy microbes.

Neutrophil extracellular traps (NETs)

- Web-like structures of DNA (chromatin) and granule proteins released by neutrophils; they trap and kill microbes (mainly bacteria and fungi) and limit spread.
- Triggered by infectious agents and mediators (chemokines, cytokines such as interferons, complement, ROS). ROS activate enzymes that convert arginine to citrulline, so chromatin unwinds; MPO and elastase enter the nucleus and loosen DNA further; the nuclear membrane ruptures and chromatin plus antimicrobial proteins are released. The neutrophil dies (a death distinct from apoptosis).



NETs: A. healthy neutrophils; B. release of nuclear material forming traps; C. electron micrograph of staphylococci trapped in NETs

Leukocyte-mediated tissue injury

- Leukocytes can damage normal tissue when activated inappropriately or excessively: (1) normal defence reactions causing collateral damage (TB, viral infections); (2) autoimmune diseases; (3) allergic reactions (asthma, allergies). They cannot tell microbes from normal tissue; the same weapons (enzymes, ROS, NO, cytokines) injure host cells.
- **How products reach the outside:** (a) *controlled secretion* of granule contents (normal but causes collateral damage); (b) *frustrated phagocytosis*: target too large or fixed (e.g. immune complexes on the glomerular basement membrane), so lysosomal enzymes are released outside; (c) *phagolysosome membrane damage* by some ingested material (e.g. urate crystals in gout), releasing enzymes into the cytoplasm and tissue.

Conditions with defective leukocyte function

Type	Condition	Defect / features
Genetic	Chediak–Higashi syndrome	Defective lysosomal granules; lysosomes cannot fuse with phagosomes → no phagolysosome; recurrent bacterial infections
Genetic	Chronic granulomatous disease (CGD)	X-linked or autosomal recessive; NADPH oxidase defect → no ROS; cannot kill microbes effectively; granulomas form
Genetic	Myeloperoxidase deficiency	Loss of MPO–H_2O_2 system; impaired hypochlorite (HOCl); mild susceptibility to fungal infection (Candida)
Acquired	Defective chemotaxis	Thermal injury, diabetes, malignancy, sepsis, immunodeficiency
Acquired	Defective adhesion	Diabetes, haemodialysis patients
Acquired	Defective phagocytosis / microbicidal activity	Leukaemia, anaemia, sepsis, diabetes, neonates, malnutrition

10. Termination of the acute inflammatory response

Inflammation protects but can damage tissue, so it must be tightly regulated. It declines because:

- **Removal of the stimulus:** mediators are made only while the offending agent is present.
- **Short half-life of mediators:** they are rapidly degraded.
- **Short life of neutrophils:** they live only hours in tissue and undergo apoptosis.
- **Active stop signals:** the process itself produces anti-inflammatory signals (**lipoxins, IL-10, TGF-β**).

11. Chemical mediators of inflammation

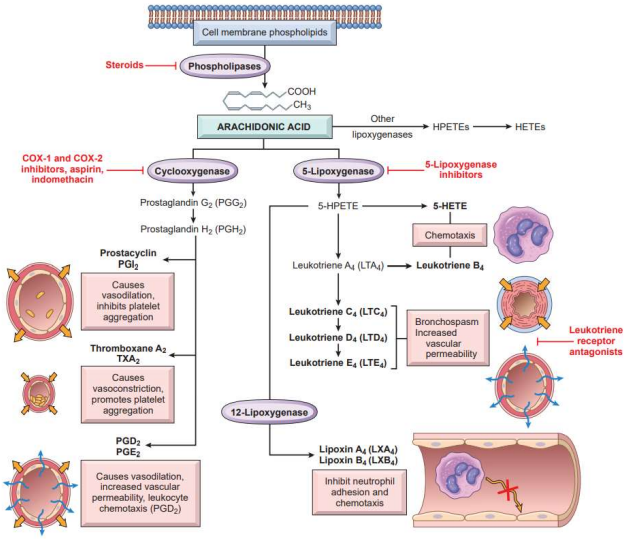
- Mediators start, amplify and regulate inflammation. Key groups: **vasoactive amines** (histamine, serotonin), **lipid mediators** (prostaglandins, leukotrienes), **cytokines and chemokines** (TNF, IL-1, IL-6, IL-8), **complement fragments** (C3a, C5a = "anaphylatoxins"), plus PAF, kinins, neuropeptides, coagulation products.
- **Cell-derived:** either *stored in granules and released rapidly* (histamine in mast cells) or *synthesised on demand* (slower; prostaglandins, leukotrienes, cytokines). Main sources: macrophages, dendritic cells, mast cells (also platelets, neutrophils, endothelium).
- **Plasma-derived:** made by the liver, circulate as inactive precursors, activated by proteolytic cascades: complement proteins (C3a, C5a, C3b), coagulation factors, kinin system (bradykinin).
- **Rules:** produced only when needed (after microbial products or necrotic cell substances); most are short-lived (broken down, inactivated or removed); one mediator can trigger others (complement fragments stimulate histamine release; TNF stimulates IL-1 and chemokines), which amplifies the response.

(i) Vasoactive amines

	Histamine	Serotonin (5-HT)
Source	Mast cells (also basophils, platelets); pre-formed in granules so released early	Platelets (released on aggregation, e.g. clotting); neuroendocrine cells of the GIT

	Histamine	Serotonin (5-HT)
Effects	Arteriolar vasodilation (redness, heat); ↑ venular permeability via **H1 receptors** → endothelial contraction → oedema. Principal mediator of the immediate transient phase of leakage. Also contracts bronchial smooth muscle (wheeze in asthma).	Vasoconstriction; minor role in inflammation. Mainly a neurotransmitter in GI tract and CNS.
Triggers	Physical injury (trauma, heat, cold); allergy (antigen binding IgE on mast cells); complement C3a, C5a; neuropeptides (substance P); cytokines (IL-1, IL-8)	Platelet aggregation
Clinical	Antihistamines block H1 receptors (urticaria, allergy)	—

(ii) Arachidonic acid (AA) metabolites



Production of AA metabolites and their roles; red X marks where drugs block (COX inhibitors, 5-LOX inhibitor, leukotriene receptor antagonists, corticosteroids)

Pathway	Products	Actions
Cyclooxygenase (COX): COX-1 (constitutive, housekeeping), COX-2 (induced in inflammation)	**PGE2**; PGD2 and PGF2α; **PGI2** (prostacyclin); **Thromboxane A2**	PGE2: fever, pain, vasodilation. PGD2/PGF2α: vasodilation, ↑ permeability. PGI2: vasodilation, inhibits platelets. TxA2 (platelets): vasoconstriction + platelet aggregation.
Lipoxygenase (LOX): leukotrienes (pro-inflammatory)	LTB4; LTC4, LTD4, LTE4 (neutrophils, macrophages, mast cells)	LTB4: strong neutrophil chemoattractant. LTC4/D4/E4: bronchoconstriction, ↑ vascular permeability, vasoconstriction.
LOX: lipoxins (anti-inflammatory)	Lipoxin A4, B4 (neutrophils, platelets; 12/15-LOX)	Inhibit neutrophil recruitment; promote resolution.

Drugs that block eicosanoids

Drug class	Examples	Action / use
COX inhibitors (NSAIDs)	Aspirin (irreversible), ibuprofen, naproxen	Block COX-1 and COX-2 → ↓ prostaglandins → ↓ pain, ↓ fever, ↓ inflammation. Do **not** reduce leukotrienes.
5-LOX inhibitor	Zileuton	Blocks leukotriene synthesis; asthma
Leukotriene receptor antagonists	Montelukast, zafirlukast	Block LTC4/LTD4 receptors; asthma, allergic rhinitis
Corticosteroids	Prednisolone, dexamethasone	↓ Phospholipase A2 (↓ AA release), ↓ COX-2, ↓ IL-1/TNF, ↓ iNOS and other inflammatory genes; very powerful; autoimmune disease, severe asthma, allergic reactions
Dietary lipids	Fish oil (omega-3)	Poor COX/LOX substrates → less inflammatory eicosanoids; promote anti-inflammatory mediators (resolvins); adjunct in chronic inflammation

(iii) Cytokines and chemokines

Cytokines are small protein messengers made mainly by activated lymphocytes, macrophages and dendritic cells (also endothelial, epithelial and connective tissue cells). They coordinate inflammation, recruit and activate leukocytes, regulate the immune response and influence healing and repair.

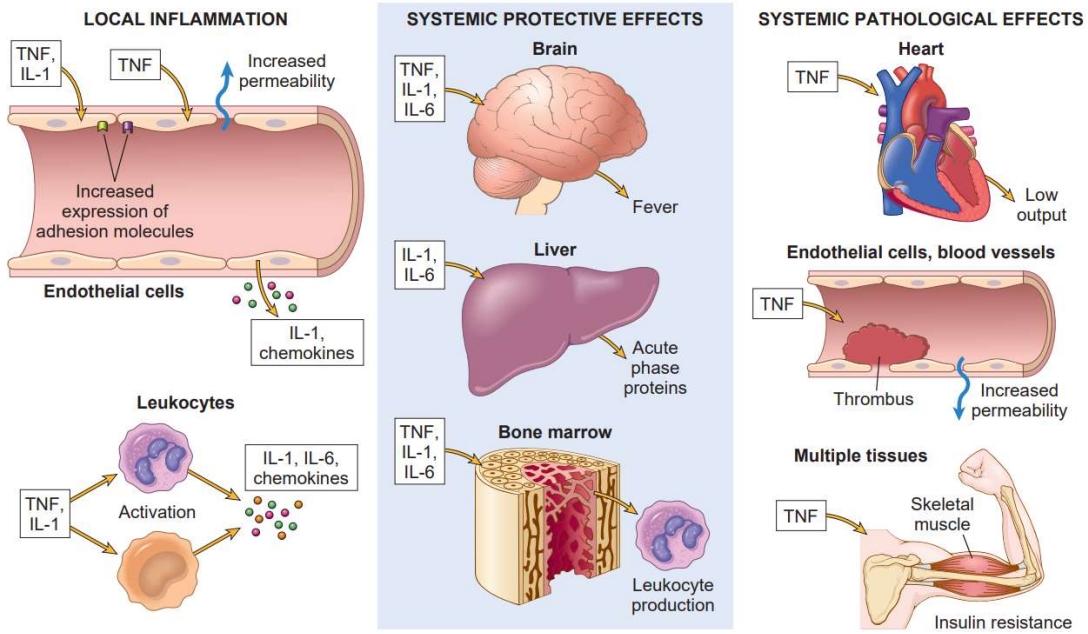
	TNF and IL-1 (key cytokines)
Source and triggers	Mainly activated macrophages and dendritic cells (TNF also T cells, mast cells; IL-1 also epithelial cells). Triggered by microbial products (via TLRs), dead cells, immune complexes, foreign bodies, physical injury. IL-1 activation needs the inflammasome.
1. Endothelial activation (most important)	↑ adhesion molecules (E-selectin, P-selectin, ICAM-1, VCAM-1), ↑ chemokines and cytokines, ↑ eicosanoids (PGs, LTs), ↑ procoagulant activity → leukocyte adhesion, migration, inflammation.
2. Activation of leukocytes and local cells	TNF ↑ neutrophil responses and activates macrophages (↑ NO). IL-1 stimulates fibroblast collagen synthesis and Th17 responses.
3. Systemic acute-phase response	TNF, IL-1, IL-6 → fever, acute-phase proteins, leukocytosis. TNF also ↑ lipid/protein breakdown, ↓ appetite; chronic TNF → cachexia (chronic infections, cancer).
Clinical correlations	**TNF antagonists** are effective in rheumatoid arthritis, psoriasis, inflammatory bowel disease, but ↑ risk of TB and intracellular infection (macrophages need TNF to kill them). IL-1 antagonists are less effective. Blocking TNF/IL-1 does not improve survival in sepsis (many other cytokines involved).

Chemokine class	Examples	Targets
CXC (α)	**IL-8** (from macrophages and endothelium)	Mainly **neutrophils**: activation + chemotaxis
CC (β)	MCP-1, eotaxin, MIP-1α, RANTES	Monocytes, eosinophils, basophils, lymphocytes; **not** neutrophils
C (γ)	Lymphotactin	Lymphocytes
CX3C	Fractalkine (membrane-bound and soluble)	Monocytes and T cells; mediates adhesion

Functions of chemokines: in acute inflammation they are made in response to microbes, necrotic cells and cytokines; they ↑ integrin affinity (stronger adhesion) and guide leukocytes by chemotaxis. *Homeostatic* chemokines organise tissue architecture (arrange T and B cells in lymph nodes and spleen).

Cytokine	Principal Sources	Principal Actions in Inflammation
In Acute Inflammation		
TNF	Macrophages, mast cells, T lymphocytes	Stimulates expression of endothelial adhesion molecules and secretion of other cytokines; systemic effects
IL-1	Macrophages, endothelial cells, some epithelial cells	Similar to TNF; greater role in fever
IL-6	Macrophages, other cells	Systemic effects (acute phase response)
Chemokines	Macrophages, endothelial cells, T lymphocytes, mast cells, other cell types	Recruitment of leukocytes to sites of inflammation; migration of cells in normal tissues
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes
In Chronic Inflammation		
IL-12	Dendritic cells, macrophages	Increased production of IFN-γ
IFN-γ	T lymphocytes, NK cells	Activation of macrophages (increased ability to kill microbes and tumor cells)
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes

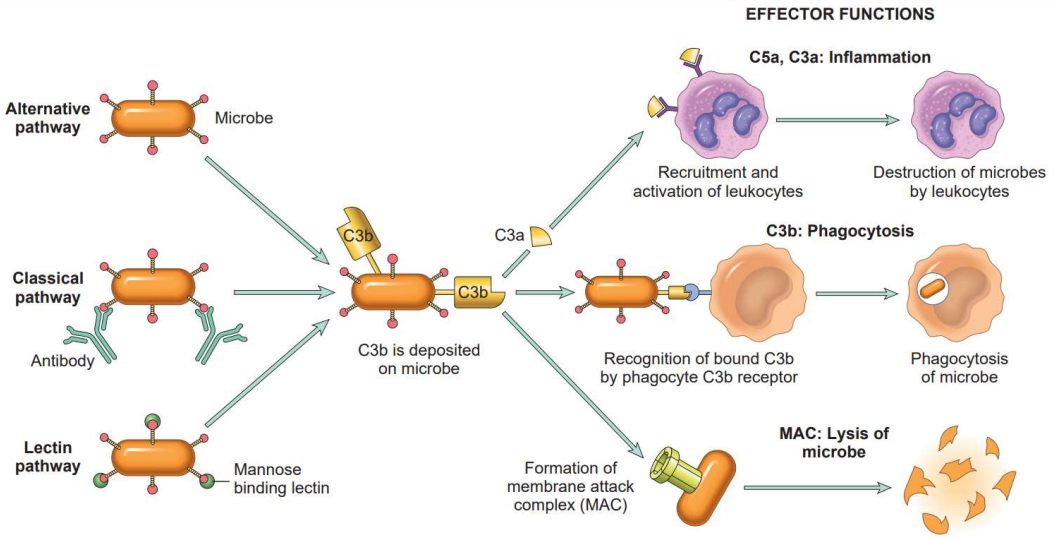
Cytokines in inflammation (slide table)



Major roles of cytokines in acute inflammation: local effects (endothelium, leukocytes) and systemic effects (fever, acute-phase proteins, leukocyte production)

(iv) Complement system

- Plasma proteins (made mainly in the liver, circulate inactive) that defend against microbes and contribute to inflammation. Activation = enzymatic cascade with strong amplification.
- **Three pathways, all producing C3 convertase (cleaves C3):** *Classical* (antigen–antibody complexes IgM/IgG; C1 binds antibody); *Alternative* (microbial surface molecules such as LPS/endotoxin; no antibody needed); *Lectin* (mannose-binding lectin binds microbial carbohydrates; activates C1-like proteins).
- **C3 cleavage:** C3 → C3a (inflammatory mediator) + C3b (binds microbe = opsonin). C3b helps form C5 convertase → C5a (inflammatory) + C5b (starts MAC).



Activation and functions of the complement system

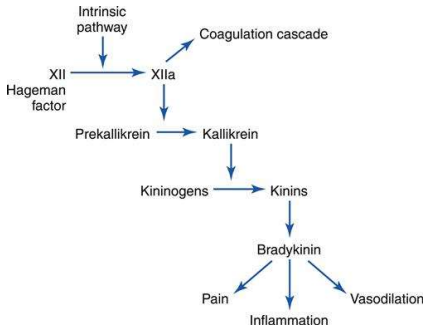
Function	Details
1. Inflammation	C3a, C4a, C5a (anaphylatoxins) → mast cell degranulation and ↑ vascular permeability. **C5a** = strong chemotactic factor for neutrophils, monocytes, eosinophils; it also activates AA metabolism → more mediators.
2. Opsonisation	C3b and iC3b coat microbes; phagocytes with C3b receptors bind and engulf more easily.
3. Cell lysis	C5b–C9 = **membrane attack complex (MAC)** → pores → osmotic lysis; important for *Neisseria* (MAC deficiency → recurrent Neisseria infections).

Regulators (protect host)	Deficiency causes
C1 inhibitor (C1-INH): blocks classical pathway	Hereditary angioedema
DAF (stops C3 convertase formation) and CD59 (stops MAC assembly), GPI-anchored	GPI-anchor deficiency → excess complement activation → paroxysmal nocturnal haemoglobinuria (PNH)
Factor H: inhibits alternative pathway, breaks down C3b	Mutation → atypical haemolytic uraemic syndrome (HUS)

Clinical relevance: excess activation in autoimmune disease (antibodies deposit on host tissue → complement → inflammation and damage). Deficiencies: early components (C1, C4) → immune complex disease; C3 → severe recurrent infections; MAC (C5–C9) → Neisseria infections.

(v) Other mediators

Mediator	Source	Actions
PAF (platelet-activating factor)	Platelets, basophils, mast cells, neutrophils, macrophages, endothelium	Originally named for platelet aggregation. Vasodilation; very potent ↑ vascular permeability (may exceed histamine); bronchoconstriction; leukocyte and platelet activation.
Coagulation products	Clotting factors, thrombin, fibrin split products	Protease-activated receptors (PARs) on many cells activated by thrombin; platelet activation. Fibrinopeptides ↑ permeability and attract leukocytes. Clotting and inflammation occur together.
Kinins / bradykinin	Plasma kininogens cleaved by kallikrein	↑ Permeability, vasodilation, smooth muscle contraction, **pain** (like histamine). Short-lived (broken down by kininase).
Neuropeptides (substance P, neurokinin A)	Sensory nerves and leukocytes; abundant in lungs and GIT	Substance P: transmits pain, ↑ permeability, neurogenic inflammation. Neurokinin A acts similarly. Leukocytes have receptors for neuropeptides (nerve–immune cross-talk); vagus nerve activation ↓ TNF release ("anti-inflammatory reflex"; trials in rheumatoid arthritis).



Kinin pathway: XII → XIIa → prekallikrein → kallikrein → kininogens → kinins → bradykinin → pain, inflammation, vasodilation

12. Morphological patterns of acute inflammation

Hallmarks of acute inflammation: dilated small blood vessels and accumulation of leukocytes and fluid in extravascular tissue. Depending on cause, severity and tissue, extra patterns appear on top; recognising them gives clues to the cause.

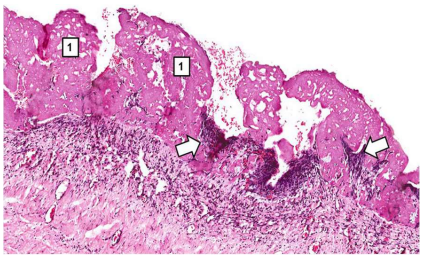
Pattern	Description	Example
1. Serous	Cell-poor, clear fluid from plasma leakage (↑ permeability) or mesothelial secretion; in spaces or body cavities (pleura, peritoneum, pericardium). Not rich in cells or microbes.	Skin blister (burn, viral infection): fluid beneath damaged epidermis
2. Fibrinous	Larger permeability increase lets fibrinogen leak out and form fibrin, especially on cavity linings (pleura, pericardium, meninges). Fibrin = pink (eosinophilic) meshwork. Removed by fibrinolysis/macrophages; if not, organised into scar, causing fibrous thickening and possible obliteration of the pericardial space.	Fibrinous pericarditis ("bread and butter" appearance)
3. Purulent (suppurative)	Pus = neutrophils, necrotic debris, oedema fluid. Usually pyogenic bacteria (e.g. staphylococci) causing **liquefactive necrosis**.	Acute suppurative appendicitis; bronchopneumonia
Abscess	Localised pus collection in tissue/organ/confined space from deep seeding of pyogenic bacteria. Centre: liquefied dead neutrophils and tissue; surrounding zone of viable neutrophils; outer rim of dilated vessels, inflammatory cells and fibroblasts (early repair). May be walled off and replaced by scar.	Lung abscesses in bronchopneumonia
4. Ulcer	Local loss of surface tissue by sloughing of necrotic inflamed tissue; only on surface-lined structures (skin, mucosa: mouth, stomach, intestine, genitourinary tract, lower limbs in poor blood supply: diabetes, PVD, sickle cell). *Acute:* neutrophils and vascular dilation at edges. *Chronic:* fibroblast proliferation, scarring, lymphocytes, macrophages, plasma cells (acute + chronic coexist).	Peptic ulcer (stomach/duodenum)



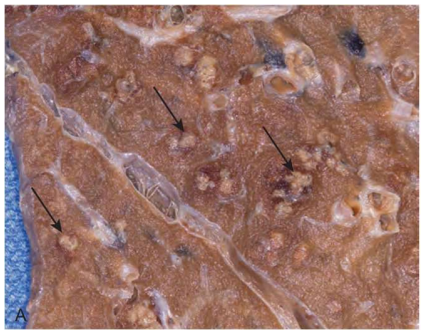
Serous: skin blister



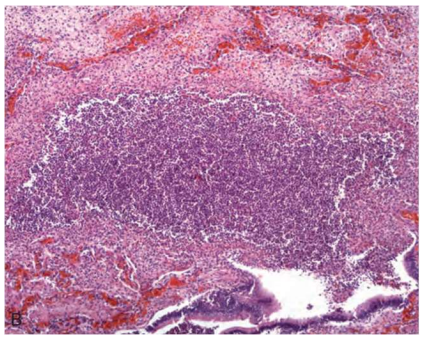
Fibrinous: pericardium (bread and butter)



Fibrinous: pink fibrin meshwork on pericardium (1)



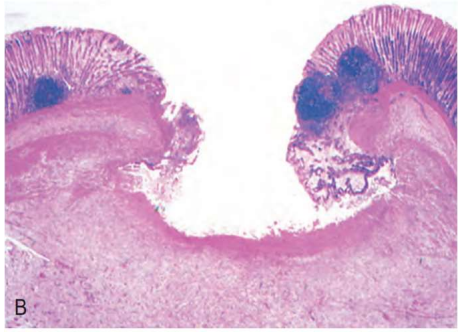
Abscesses (arrows) in lung, bronchopneumonia



Abscess: neutrophils and debris surrounded by congested vessels



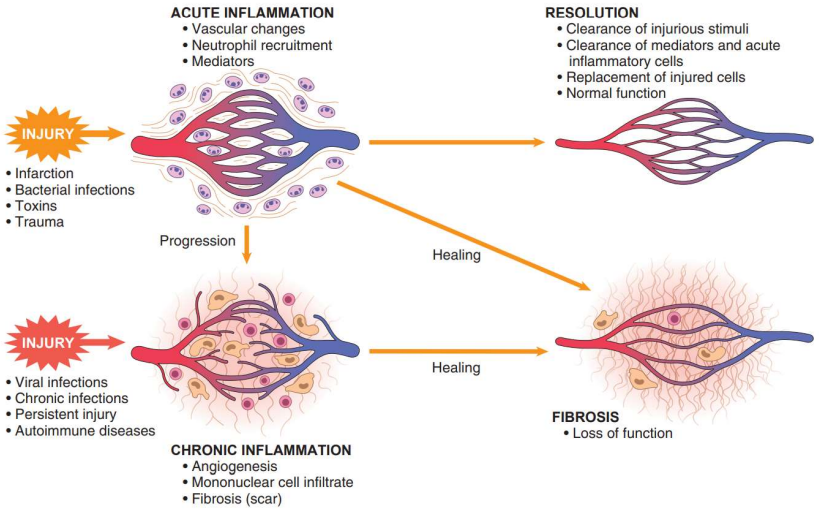
Chronic duodenal ulcer



Duodenal ulcer crater with acute inflammatory exudate in the base

13. Outcomes of acute inflammation and factors that modify it

Outcome	When	What happens
1. Complete resolution	Injury short-lived, mild, minimal destruction; tissue can regenerate	Macrophages remove dead cells and microbes; oedema drains via lymphatics; tissue regenerates to normal
2. Healing by fibrosis (scar)	Extensive destruction; tissue cannot regenerate; fibrin exudate cannot be cleared (e.g. pleura, peritoneum)	Fibroblasts grow in = **organisation** → fibrous scar
3. Progression to chronic inflammation	Persistent infection/stimulus or interference with normal healing	Chronic inflammation with ongoing tissue injury and repair



Outcomes: resolution, healing by fibrosis, chronic inflammation

Factors modifying inflammation	Details
Related to the organism	Type, intensity and site of injury; virulence and load of the infective agent; portal of entry of microbes
Related to the host	General health (malnutrition, diabetes, alcoholism ↑ susceptibility); immune status (immunodeficiency, neutropenia); local factors (ischaemia, necrosis, foreign bodies); tissue type (some regenerate, some cannot)

14. Summary of acute inflammation (lecturer's summary)

Acute inflammation is an immediate protective response to microbes, dead cells or injury. Aim: eliminate the cause, remove damaged tissue, start repair.

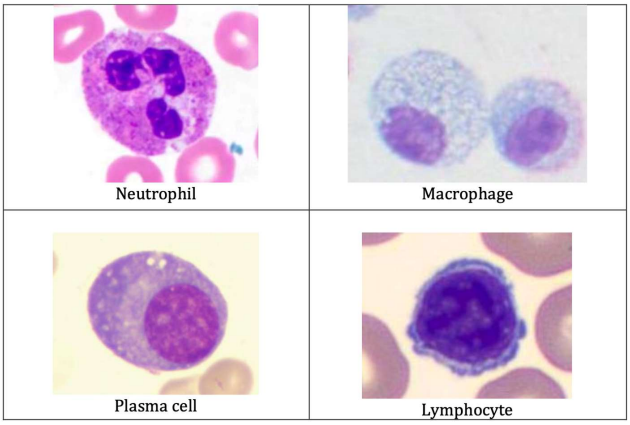
Section	Key points
Vascular changes and cardinal signs	Vasodilation → ↑ blood flow → redness, heat. ↑ permeability → protein-rich oedema → swelling. Mediators (PGs, bradykinin, neuropeptides) → pain. Pain + tissue injury → loss of function.
Cellular events	Neutrophils arrive first: adhesion, transmigration, chemotaxis. Activation by microbes and mediators. Phagocytosis: recognition → engulfment → phagolysosome killing. Killing by ROS, NO, lysosomal enzymes (may also damage host tissue).
Key mediators	Histamine, serotonin; prostaglandins, leukotrienes, lipoxins (AA pathway); cytokines (TNF, IL-1, IL-6), chemokines (IL-8); complement (C3a, C5a, C3b, MAC); PAF, bradykinin, neuropeptides.
Outcomes	Resolution (minimal damage, regeneration); scar (severe injury/non-regenerative tissue); chronic inflammation (persistent stimuli).

PART 2 · PRACTICAL: ACUTE INFLAMMATION

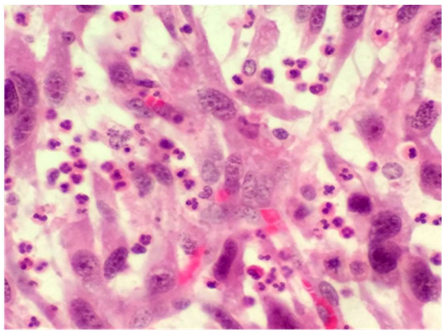
Objective: identify the gross and microscopic morphology of acute inflammation in **pneumonia** (lung) and **acute appendicitis**. Answers below come from the lecture content and the pictures; confirm numbered labels with your lecturer.

A. Identify inflammatory cells (pictures A–D)

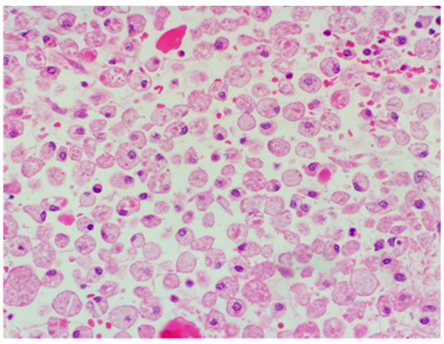
Task: study pictures A, B, C, D; identify the cells; state the conditions/disease processes where each white blood cell type is predominantly seen.



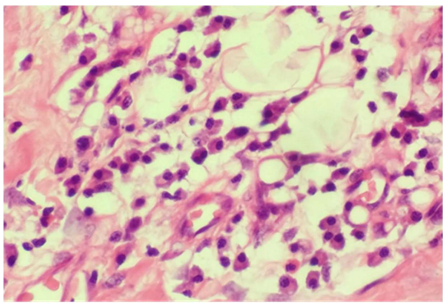
Reference cells: neutrophil, macrophage, plasma cell, lymphocyte



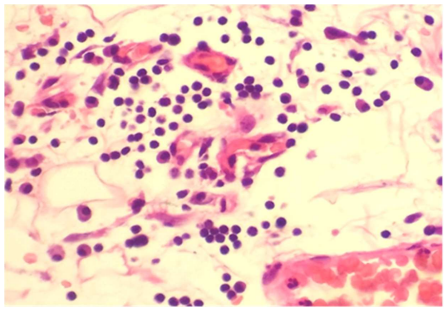
A



B



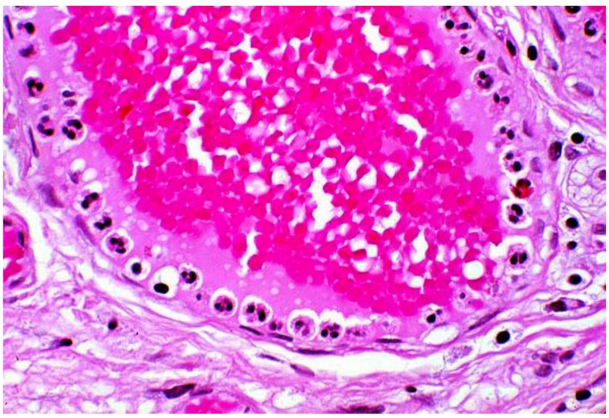
C



D

Picture	Cell	How to recognise (H&E)	Predominant in
A	**Neutrophils**	Small cells with multilobed (2–5 lobes) dark nucleus, pale pink cytoplasm (small dark dots scattered between larger tissue cells)	Acute inflammation: bacterial infection, pus, tissue necrosis
B	**Macrophages**	Large cells, abundant pale/foamy cytoplasm, round or oval nucleus	Late acute and chronic inflammation, granulomas (e.g. TB), clearing debris, repair
C	**Plasma cells**	Round/oval cell, eccentric nucleus with "clock-face" chromatin, basophilic cytoplasm with clear perinuclear halo	Chronic inflammation (antibody production)
D	**Lymphocytes**	Small round cells, dense dark round nucleus, very thin rim of cytoplasm	Chronic inflammation, viral infections, autoimmune disease

B. Stage of cellular event (picture E)



E: a venule with WBCs lining the wall and passing through it

The 3 stages of cellular events present in E: **(1) margination (and rolling)**, **(2) adhesion to the endothelium**, **(3) transmigration/diapedesis** through the vessel wall into the tissue. (The red cells fill the vessel lumen; the neutrophils sit along the wall.)

Case study 1: Acute inflammation in bronchopneumonia (F–J)

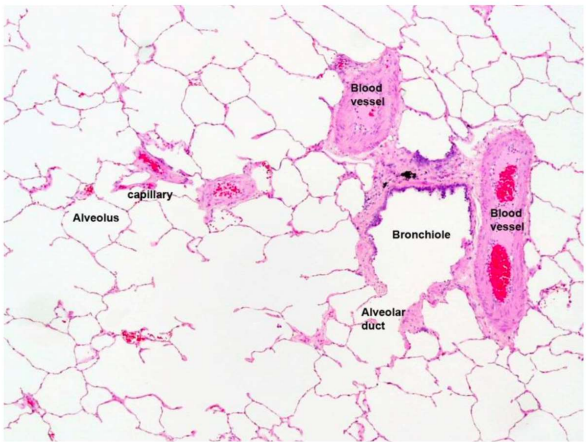


F: lung with pneumonia. Arrows = consolidated areas

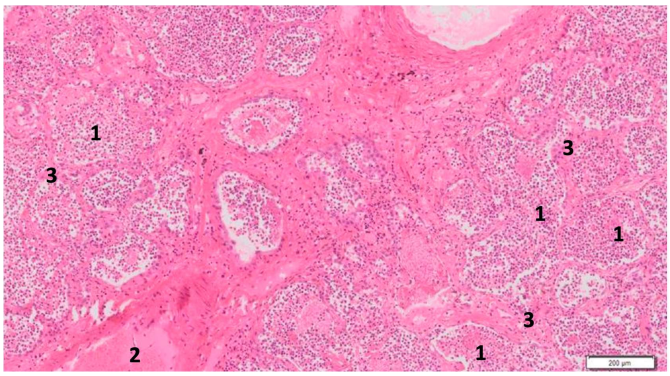


G: cut surface of normal lung

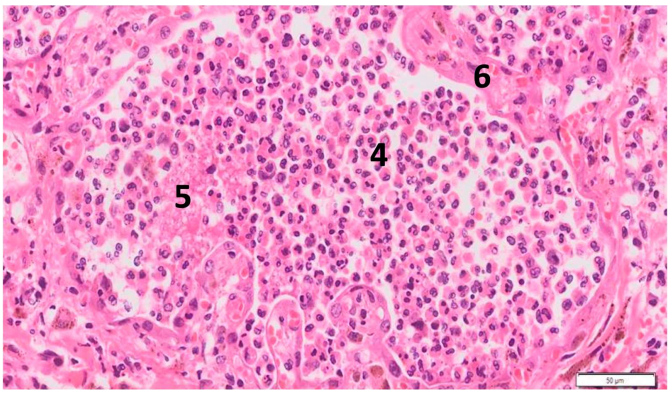
F vs G (gross): normal lung (G) is spongy, uniform brown-pink and air-filled. Pneumonia (F) has patchy **consolidated** areas (arrows): firm, solid, pale yellow-tan, no longer air-filled, centred around bronchi (bronchopneumonia pattern).



H: normal lung histology (alveoli, capillaries, alveolar duct, bronchiole, blood vessels)



I: bronchopneumonia, low power (labels 1–3)



J: bronchopneumonia, high power (labels 4–6)

Tasks (H normal vs I, J bronchopneumonia):

- **1. Features of acute inflammatory exudate:** alveoli filled with neutrophils plus pink oedema fluid (and fibrin) instead of air; congested capillaries in the alveolar walls.
- **2. Vessels and neutrophils:** blood vessels are congested/dilated, with neutrophils lining the wall (margination) and moving out of the vessel into surrounding tissue and alveoli (emigration).
- **3. Normal lung:** few neutrophils, located inside capillaries; alveolar spaces are empty/air-filled; thin alveolar walls.

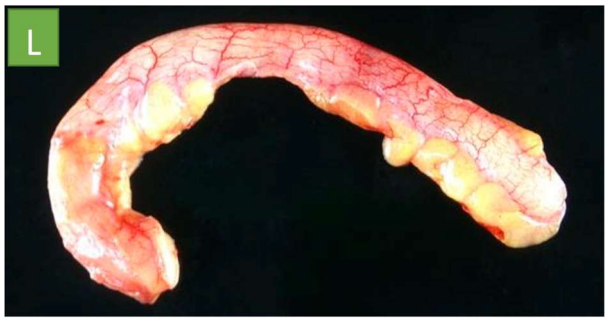
No.	Likely identity (best reading of the picture)
1	Alveoli filled with inflammatory exudate (neutrophils)
2	Larger blood vessel / bronchial wall structure
3	Thickened, congested alveolar septa between the filled alveoli
4	Neutrophils (polymorphs) in the alveolar lumen
5	Pink proteinaceous oedema fluid/fibrin with cell debris in the alveolar lumen
6	Alveolar wall with congested capillaries

Case study 2: Acute appendicitis (K–P)

Case: 12-year-old boy with fever, nausea and abdominal pain for 2 days. Pain began around the umbilicus and shifted to the right lower abdomen; tender over the right iliac fossa. Appendix removed. (Pain migrates because early visceral pain near the umbilicus becomes localised pain when the inflamed appendix irritates the peritoneum.)

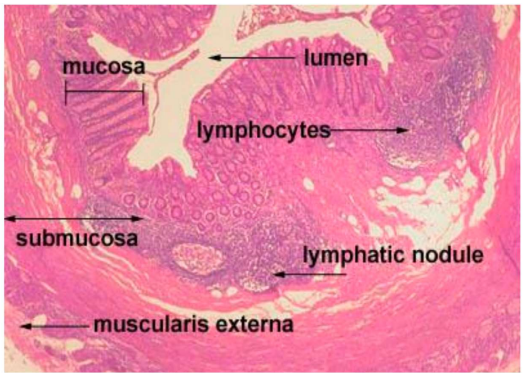


K: patient's appendix



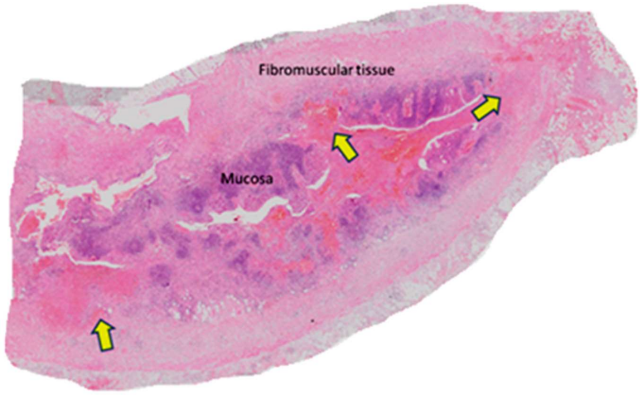
L: normal appendix

Gross pathology in K (3 features of acute inflammation): (i) congested, dilated serosal blood vessels (redness); (ii) swollen, enlarged, thickened appendix (oedema); (iii) dull yellow-green purulent/fibrinous exudate on the surface (compare the slim, pale pink normal appendix L).

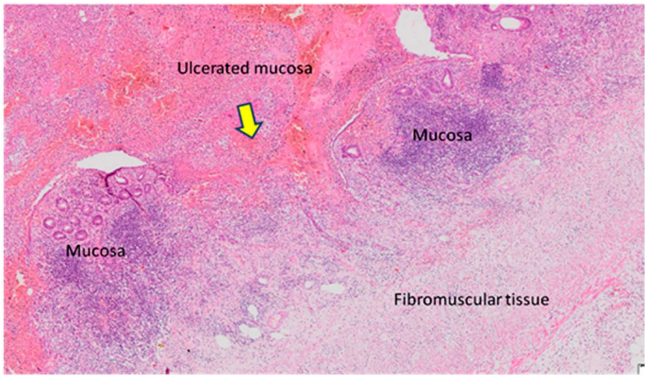


M

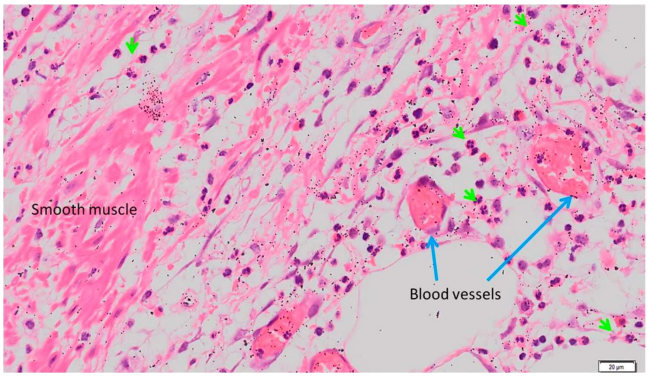
M: normal appendix: intact mucosa, submucosal lymphoid nodules, muscularis externa



N: boy's appendix (low power): mucosa, fibromuscular tissue, arrows



O: ulcerated mucosa (arrow), mucosa, fibromuscular tissue



P: smooth muscle, blood vessels, neutrophils (green arrows)

Microscopic pathology in N, O, P (4 features): (i) **mucosal ulceration**: lining lost (ulcerated mucosa); (ii) **inflammatory cells**: dense neutrophil infiltrate through mucosa, submucosa and the muscle wall (green arrows in P); (iii) **necrosis** of the wall (pink, structureless areas); (iv) **vascular changes**: congested dilated vessels (blue arrows in P) and oedema between smooth muscle fibres. Together = **acute suppurative appendicitis**.

PART 3 · FLIPPED CASE: DARLA (cat bite)

Case: Darla, 25, swollen right middle finger after a cat bite. Within 24 hours the skin around the puncture wound reddened and the tissue swelled; her PIP joint became extremely painful and she partially lost the ability to flex the finger. Temperature 38 °C; finger swollen; skin at the wound warmer than the same site on the opposite hand. The bite site is most likely contaminated by bacteria. She was given medication to relieve symptoms, and antibiotics.

The slides give the questions only; these are suggested answers from the lecture content. Check with your lecturer.

1. State the pathological process

Acute inflammation (acute bacterial, suppurative inflammation of the finger after a contaminated puncture wound; pain at the PIP joint suggests possible spread to the joint).

2. Define it

A rapid, protective response of vascularised tissue to infection or injury that delivers leukocytes and plasma proteins to the site to eliminate the cause, remove damaged tissue and start repair; develops in minutes to hours and lasts hours to a few days.

3. Cardinal signs in her hand

Redness (skin reddened), swelling (tumor), heat (warmer than the other hand), pain (extremely painful PIP joint), loss of function (partial loss of flexion). Plus systemic fever (38 °C).

4. Pathophysiology of the signs

Redness and heat: histamine relaxes arterioles → ↑ blood flow into capillaries/venules. **Swelling:** histamine and other mediators ↑ venular permeability → protein-rich fluid leaks → oedema. **Pain:** PGE2 sensitises nerve endings to bradykinin and other pain mediators (swelling also stretches tissue). **Loss of function:** pain + swelling + tissue injury. **Fever:** IL-1, TNF and IL-6 act on the hypothalamus via PGE2.

5. Pathophysiologic categories of oedema

(1) ↑ hydrostatic pressure (heart failure, venous obstruction); (2) ↓ plasma osmotic pressure (low albumin: liver disease, nephrotic syndrome, malnutrition); (3) Na^+ and water retention (renal dysfunction, salt); (4) lymphatic obstruction (tumour, fibrosis, surgery, infection); (5) **inflammation** (↑ vascular permeability; protein-rich exudate). Darla's oedema is category 5, an **exudate**.

6. Causes that stimulate this process

Infection (the main cause here: bacteria introduced by the cat bite), tissue necrosis/trauma (puncture wound), foreign bodies, immune reactions. Outcome depends on pathogen, host response and host factors.

7. Vascular and cellular events

Vascular: transient vasoconstriction → vasodilation (↑ flow) → ↑ venular permeability (endothelial contraction by histamine/bradykinin/leukotrienes; injury) → exudate/oedema → stasis. **Cellular:** margination → rolling (selectins) → firm adhesion (integrins) → transmigration (CD31, collagenases) → chemotaxis (bacterial products, IL-8, C5a, LTB4) → activation → phagocytosis. Neutrophils within 6–24 h, then macrophages (24–48 h).

8. Chemical mediators and actions

Histamine (vasodilation, ↑ permeability); serotonin (minor); PGE2 (fever, pain, vasodilation), PGD2/PGF2α (vasodilation, ↑ permeability), thromboxane; LTB4 (neutrophil chemotaxis), LTC4/D4/E4 (↑ permeability); TNF, IL-1 (endothelial activation, fever), IL-6 (acute-phase); IL-8 (neutrophil chemotaxis); complement C3a/C5a (↑ permeability, chemotaxis), C3b (opsonin), MAC (lysis); bradykinin (pain, ↑ permeability); PAF; neuropeptides (substance P); lipoxins (stop signal).

9. Phagocytosis of the bacteria

(1) **Recognition and attachment:** phagocyte receptors (mannose, scavenger, and receptors for opsonins IgG and C3b) bind the bacteria. (2) **Engulfment:** membrane wraps around the bacterium → phagosome → fuses with lysosome → phagolysosome. (3) **Killing and degradation:** ROS, MPO–H_2O_2–HOCl system, NO, and lysosomal enzymes. Neutrophils may also release NETs. Leakage of these products can damage nearby tissue.

10. Most likely morphological pattern

Purulent (suppurative) inflammation (pus: neutrophils, necrotic debris, oedema fluid) from pyogenic bacteria; may progress to an **abscess**.

11. Possible outcomes

Complete resolution (likely with prompt antibiotics; tissue regenerates); **healing by fibrosis** (scar and stiff finger if tissue destruction or abscess); **progression to chronic inflammation** if bacteria persist. Possible spread: lymphangitis, lymphadenitis, septic arthritis of the PIP joint.

12. Prescribed medications, with reasons

Antibiotics: kill the bacteria, removing the cause (for cat bites, a combination such as amoxicillin-clavulanate is commonly used; use the example your lecturer gives). **Symptom relief:** an NSAID (e.g. ibuprofen) blocks COX-1/COX-2 → ↓ prostaglandins → ↓ pain, fever and swelling; paracetamol is an alternative for pain and fever.

PART 4 · ONE-PAGE REVISION

Topic	Remember this
Definition and aim	Protective vascular response; eliminate cause, remove damaged tissue, start repair
5 Rs	Recognition, Recruitment, Removal, Regulation, Repair
5 cardinal signs	Rubor, tumor, calor, dolor, functio laesa (Celsus 4; Virchow added the 5th)
Causes	Infection (most common), necrosis, foreign bodies, immune reactions
Recognition	TLRs (microbes); NLRs → inflammasome → IL-1 (cell damage); opsonin receptors; complement, MBL, collectins
Hallmark of acute	↑ Vascular permeability of venules → exudate; neutrophils
Vascular sequence	Transient vasoconstriction → vasodilation → ↑ permeability → oedema → stasis → margination
Permeability mechanisms	Endothelial contraction (histamine; immediate 15–30 min; delayed 2–12 h e.g. sunburn); endothelial injury (burns, toxins)
Exudate vs transudate	Exudate: high protein, inflammation. Transudate: low protein, pressure imbalance
Oedema categories	↑ hydrostatic; ↓ oncotic; Na/water retention; lymphatic obstruction; inflammation
Leukocyte steps	Margination → rolling (selectins) → adhesion (integrins; ICAM-1/VCAM-1) → diapedesis (CD31) → chemotaxis (IL-8, C5a, LTB4, bacterial products)
Cells over time	Neutrophils 6–24 h → macrophages 24–48 h → lymphocytes/plasma cells if chronic
Phagocytosis	Recognition (opsonins IgG, C3b, MBL) → engulfment (phagolysosome) → killing (ROS, MPO/HOCl, NO, enzymes); NETs
Leukocyte injury / defects	Controlled secretion, frustrated phagocytosis, phagolysosome damage; Chediak–Higashi, CGD (NADPH oxidase), MPO deficiency
Termination	Stimulus gone, short mediator half-life, neutrophil apoptosis, lipoxins/IL-10/TGF-β
Key mediators	Histamine (early; H1); PGE2 (pain, fever); LTB4 (chemotaxis); LTC4/D4/E4 (bronchoconstriction); TNF/IL-1 (endothelial activation, fever); IL-8; C5a, C3b, MAC; bradykinin (pain); PAF; lipoxins (stop)
Drugs	NSAIDs (COX), steroids (PLA2), antihistamines (H1), montelukast (LT receptor), zileuton (5-LOX), TNF antagonists (↑ TB risk)
Patterns	Serous, fibrinous, purulent/abscess, ulcer
Outcomes	Resolution, fibrosis (scar, organisation), chronic inflammation
Practical cells	Neutrophil (multilobed), macrophage (big, pale), plasma cell (eccentric nucleus), lymphocyte (small, round, dark)
Pneumonia	Gross: patchy consolidation. Micro: alveoli filled with neutrophils, oedema, fibrin; congested septa
Appendicitis	Gross: swollen, congested, exudate on serosa. Micro: mucosal ulcer, neutrophils through wall, necrosis, congested vessels